



ST498 Capstone Project

MSc Data Science

Department of Statistics

Dr Marcos Barreto

Associate Professor in Data Science (Education)

Capstone project co-ordinator

What is a capstone project?



Capstone project: *a capstone experience, culminating project, or senior exhibition,* among many other terms, a capstone project is **a multifaceted assignment that serves as a culminating academic and intellectual experience for students**, typically during their final year of high school or middle school, or **at the end of an academic programme or learning-pathway experience.**

... capstone projects may take a wide variety of forms, but most are long-term investigative projects that **culminate in a final product, presentation, or performance.**

Source: <https://www.edglossary.org/capstone-project/>

What is a capstone project? (cont'd)



Capstone projects are generally designed to **encourage students to think critically, solve challenging problems, and develop skills such as oral communication, public speaking, research skills, media literacy, teamwork, planning, self-sufficiency, or goal setting**—i.e. skills that will help prepare them for modern careers.

In most cases, the projects are also interdisciplinary, in the sense that they require students to **apply skills or investigate issues across many different subject areas or domains of knowledge**. Capstone projects also tend to **encourage students to connect their projects to community issues or problems**, and to integrate outside-of-school learning experiences, including activities such as interviews, scientific observations, or internships.

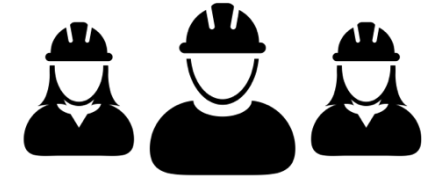
Project team



⋮



Student team
members



Partner supervisor
(and team)



LSE supervisor

Capstone project
Academic collaboration tackling
an applied data science problem

LSE supervisors (to be confirmed)



Mona Azadkia



Marcos Barreto
(course co-ordinator)



Umut Cetin



Alessandro De Palma



Piotr Fryzlewicz



Leonardo Dias



Ieva Kazlauskaite

LSE supervisors



Joshua Loftus



Gelly Mitrodima



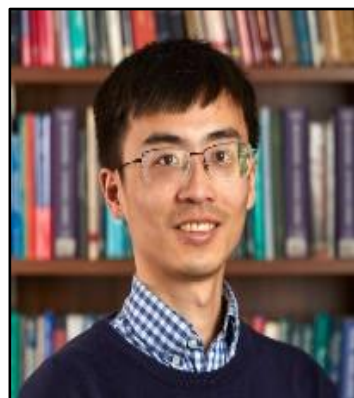
Wicher Bergsma



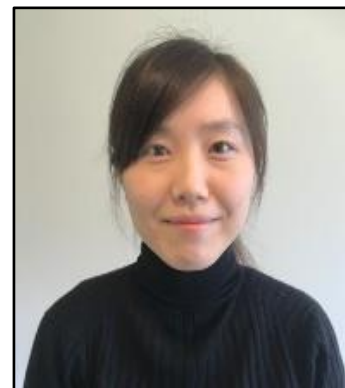
Andreas Søjmark



Zoltan Szabo



Tengyao Wang



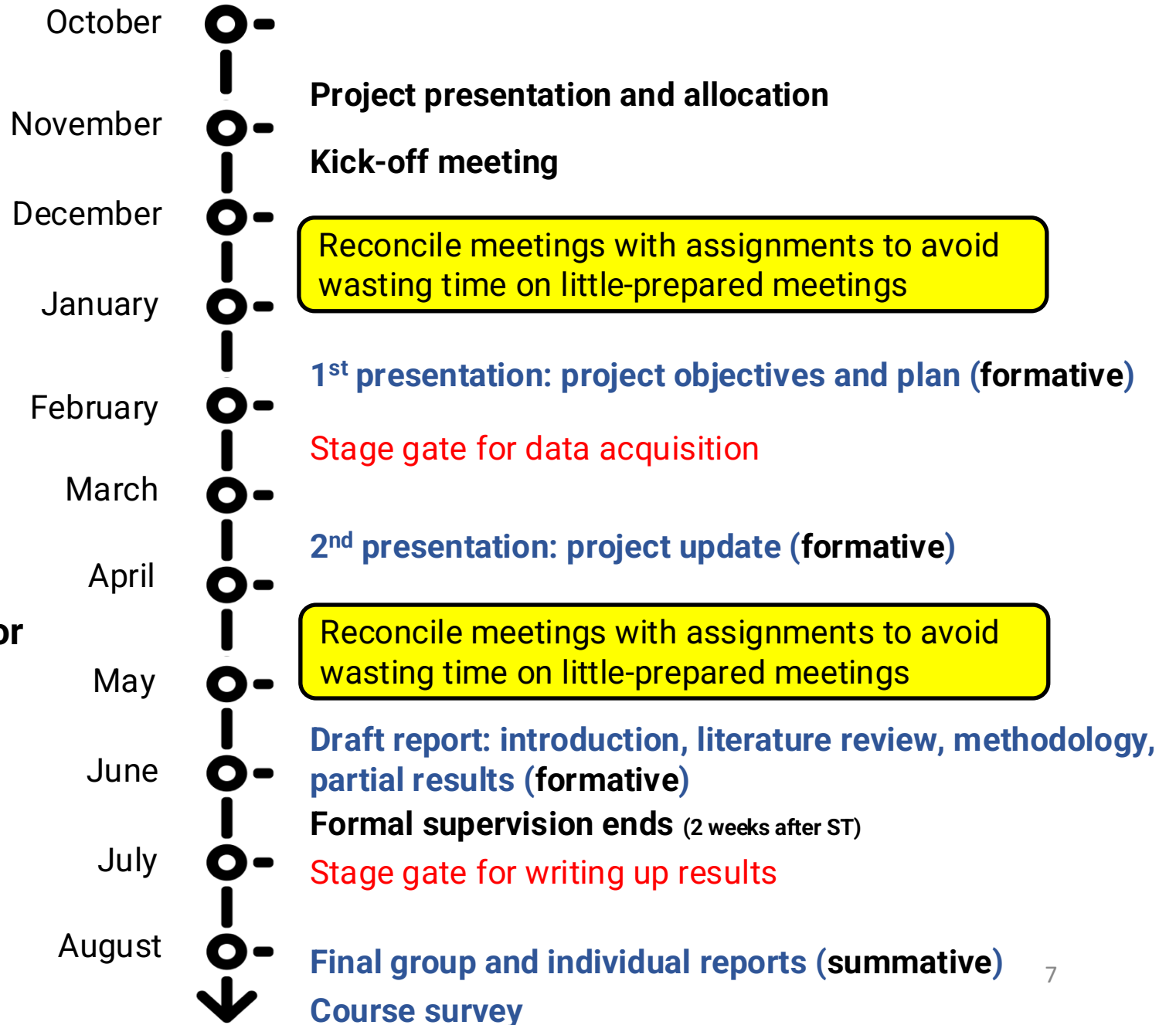
Christine Yuen



Xiaolin Zhu

Milestones

- ✓ Interaction with partner/supervisor
- ✓ Deliverables
- ✓ Important milestones
- ✓ Useful tips



ALL DATES TO BE CONFIRMED

Deliverables

- **Mandatory**

- **1st presentation:** project objectives and plan (*slides + 10 min video presentation*)
- **2nd presentation:** update on the project, including partial results (*updated slides + 10 min video presentation*)
- **Draft group report*** (*introduction, literature review, methods, partial results*)
- **Final group and individual reports*** (*updated draft report + final results, analysis and discussion*)

- **Optional**

- Publications, code/data repositories, presentations to partners etc
(to be discussed/agreed upon with the partner)

* See course guide on Moodle for the structure

Your responsibilities

- **Project management**

- Make sure to agree on a project work plan
- Make sure to add individual contributions
- Make sure to achieve various milestones and deliverables
- Maintain good relations with your peers, project partner and supervisor

- **Technical contributions**

- Partitioning the work as you appreciate it fits you best
- Pursue research, data analysis, and software development
- Being proactive in unblocking yourself with data, APIs, compute resources etc
- Project presentation and report writing



Project plan: The next intermediate steps will be extracting relevant satellite data and performing some EDA

What we agreed on the 26.03.2024

Todos/ Next steps	Responsible	Deadline
Check access to big machines at LSE	Team	12/04
Write function to extract spatial data based on lat and lon (1km ²) corresponding to water quality sample info (only subset for now) → store in memory	Team	12/04
EDA of these data for small set of points and short time frame (1 month/ 1 year) (missing values/ sparsity, averages, etc.) using pandas/ numpy, etc.	Team	12/04
Iteratively extract satellite info for all water quality sample locations and times, combine to one netCDF	Team	tbd
Join satellite with water quality data	Team	tbd
Train models of different specifications and evaluate performance	Team	
Look into Land use information to determine format/ usability		12/04

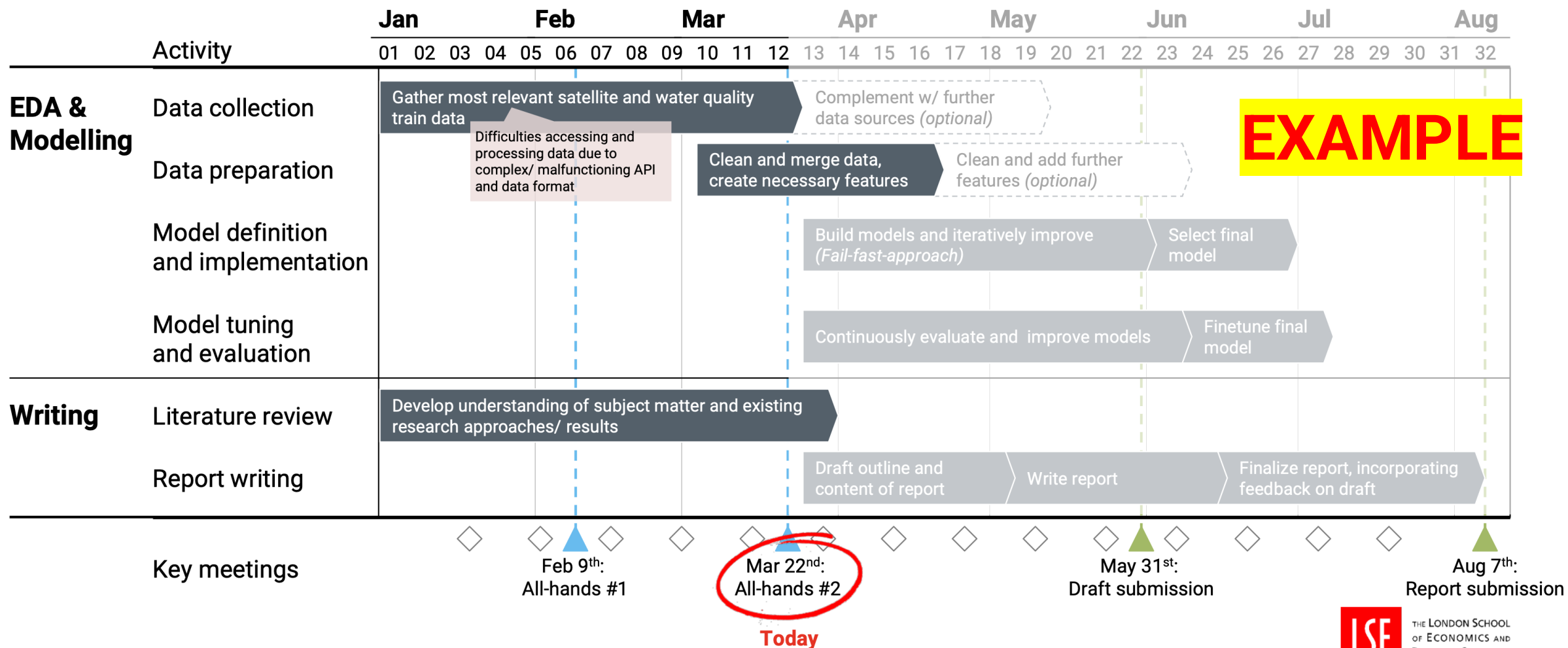
EXAMPLE



Project plan: We needed to adjust the project timeline due to complex satellite data access

DRAFT – to be discussed

◇ Bi-weekly project partner meetings ▲ Irregular course meetings ▲ Deadlines



Your responsibilities

- **Privacy and onboarding training**

- Some partners require that **Non-Disclosure Agreement** (NDA) and/or **Student Collaboration Agreement** documents be signed by everyone involved in the project. They may also ask for some **onboarding training** before providing access to any datasets. Engage with that as soon as possible!
- Regardless of NDAs, you **MUST NOT** disclose any information regarding your project without explicit authorisation by the partner – this includes posts on social media, LinkedIn profiles, CVs etc. If a potential employer asks you for a portfolio of projects, we (LSE) can provide you with an “**evidence letter**”.
- You **MUST** discuss with your partner and LSE supervisor whether you can make available any code, data, or draft/final project report.

Your responsibilities

- **Generative AI tools**

- You are **ALLOWED** to use generative AI tools to help with your writing and coding, but you **MUST** discuss with your LSE supervisor and industry partner whether the use of such tools is desirable and under what circumstances.
- You can use LLMs as “objects” of research or part of your methodology, when appropriate.
- You **MUST** acknowledge the use of generative AI tools in any outputs (presentations, reports, code etc). Refer to **Moodle for full guidance** and Department/course-specific policies.

Our responsibilities

- **LSE supervisors and course co-ordinator**
 - Ensure a fun and effective way of working together
 - Ensure the project progresses as expected
 - Ensure that students and partners are engaged and committed to deadlines and high-standard results
 - Provide academic and technical support throughout the project
 - Ensure compliance with any NDAs and School/Department/course-level policies
 - Provide access to technical resources (GitHub, Google Cloud Computing)
- **Please, speak to your supervisor to discuss any issues with your group and/or partner. Escalate to the course co-ordinator if necessary.**

Project assignments

All details to be discussed in our meeting on 19 October.

- Our goal is to have **16 groups with 3 students.**
 - It is your task to agree on group formation.
 - We can help anyone to find group members.
- Need **individual preferences**
 - **Each student** needs to send us individual preferences for project topics.
 - Individual preference: **ranking of ALL projects in decreasing order of preference.**
- Having group memberships and individual preferences as input, **groups will be assigned to projects by a maximum weight-matching procedure.**
- **Each group will be given a GitHub repository to keep data/code (if desired) and for formative and summative submissions**

Project assignments

**Actions to be taken
after the 19 October
meeting**

Action items:

- Get to know each other and decide on your group
- Look at the proposals (link on [Moodle](#))
- Group to send an [email to Marcos Barreto](#) asserting:
 - who are the group members (**name, email, LSE student ID, and GitHub username**), and
 - **for each member**, individual decreasing order of preference over **ALL** projects.
 - **Group can agree on same preference list and send just one list!**
 - Email to **include all group members in CC of the email**
 - This is to be done by **Thursday, 30 October 2026, 12:00 pm (London time)**
 - **The assignment of projects to groups will be announced within a few days from this date**, and a kick-off meeting will be arranged with partners and supervisor in early-mid November.

Project assignments

Example of preference list

Given the set of example projects:

- P1 – Applying deep learning to weather data
- P2 – Scalable data engineering methods for climate data
- P3 – Recommendation system for stock trading
- P4 – Multimodal disease classification using patient data

Your list will have the following format: (2, 4, 1, 3, ...)

Please, **DO NOT CHANGE** project numbers when sending your preference!

CONFIDENTIALITY NOTICE

THE FOLLOWING SLIDES ARE STRICTLY CONFIDENTIAL.

DO NOT DISCLOSE, PROJECT, OR PUBLISH OUTSIDE THE SCOPE OF THIS COURSE AND WITHOUT EXPLICIT AUTHORISATION OF THE COURSE CO-ORDINATOR AND/OR THE INDUSTRYPARTNER.

PROJECT BRIEFS TO BE ADDED ON 19 OCTOBER